ANALYSTLAB AFRICA

Machine Learning Internship Programme

Business Understanding Report

Week 2: Data Preprocessing & Feature Engineering

Project: ShopSmart Retail Analytics

Field	Value
Programme	Machine Learning Internship
Week	Week 2
Document	Business Understanding Report
Dataset	ShopSmart Retail (multi-table)
Prepared for	AnalystLab Africa

1. Business Objective

ShopSmart Retail Ltd operates multiple stores across Nigeria, selling products in Beauty, Electronics, Fashion, Groceries and Home categories. Management has observed fluctuations in sales, inventory pressure and rising product returns. As a Junior Machine Learning Engineer at AnalystLab Africa, the objective of this workstream is to prepare a clean, machine-learning-ready dataset that can support downstream predictive models.

Specifically, the prepared data should enable models for:

- Customer segment classification and upgrade propensity scoring
- Customer lifetime value (CLV) estimation
- Return-risk scoring at order or customer level
- Next-purchase and high-value customer propensity modelling

2. Business Context & Problem Framing

The organisation holds rich transactional and master data (customers, orders, products, stores, returns, inventory, employees). However, these tables exist in raw form with inconsistent date formats, categorical variables that models cannot consume directly, potential outliers in monetary fields, and no unified customer-level behavioural view. Without systematic preprocessing and feature engineering, any model trained on the raw tables would be unreliable.

Week 2 therefore focuses on transforming the multi-table ShopSmart data into a single, documented, machine-learning-ready customer feature matrix while recording every design decision.

3. Data Sources

Table	Rows	Role
Customers	5,000	Demographics, segment, registration date
Orders	50,000	Transactional facts (quantity, price, discount, channel, status)
Products	500	Category, brand, cost and selling price
Stores	25	Region, location, opening date
Returns	2,500	Return events, reason, refund amount
Employees	125	Store staffing (aggregated)
Inventory	500	Stock levels by product-store (aggregated)

Table 1: Source tables used in the preprocessing pipeline

4. Success Criteria for Preprocessing

- No missing values in the final modelling matrix
- Categorical variables correctly encoded (ordinal or one-hot as appropriate)
- Numerical features scaled in a way that is robust to outliers
- Extreme outliers treated (winsorized) rather than silently dropped
- Low-variance and highly collinear features removed
- Every decision documented so the pipeline is auditable and reproducible
- Final dataset exported as CSV and ready for model training

5. Key Business Questions Addressed

#	Business Question
1	What data quality issues exist in the dataset?
2	Which features require preprocessing?
3	Which encoding techniques are most appropriate?
4	Which scaling method should be applied?
5	Are there significant outliers that need treatment?
6	Which features contribute most to solving the business problem?
7	Is the dataset ready for model training?

Detailed answers to these questions are provided in the companion Data Preprocessing Report.

6. Scope and Constraints

This report and the associated notebook focus on preprocessing and feature engineering only. Model training, hyperparameter tuning and deployment are out of scope for Week 2 (they build on the foundations established in Week 1). The supplier table was reviewed and correctly excluded because it has no direct product-level foreign key usable for order enrichment.

7. Expected Outputs

- Jupyter notebook with full preprocessing pipeline
- Machine-learning-ready customer dataset (CSV)
- This Business Understanding Report (PDF)
- Data Preprocessing Report (PDF)

- Supporting visualisations (≥ 6 charts)